

REPLACEMENT PARTS MANUAL

HEAT KING HK300

Mobile Glycol Heating System





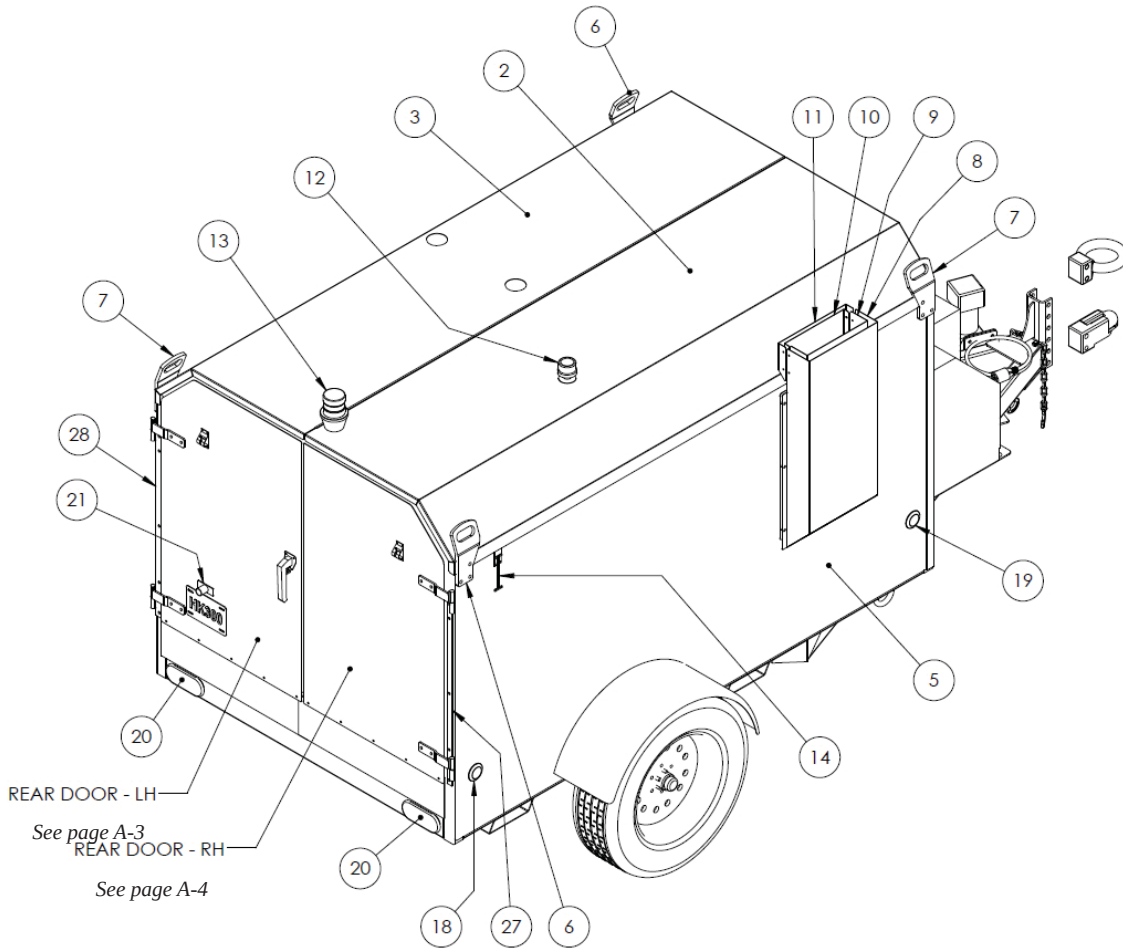
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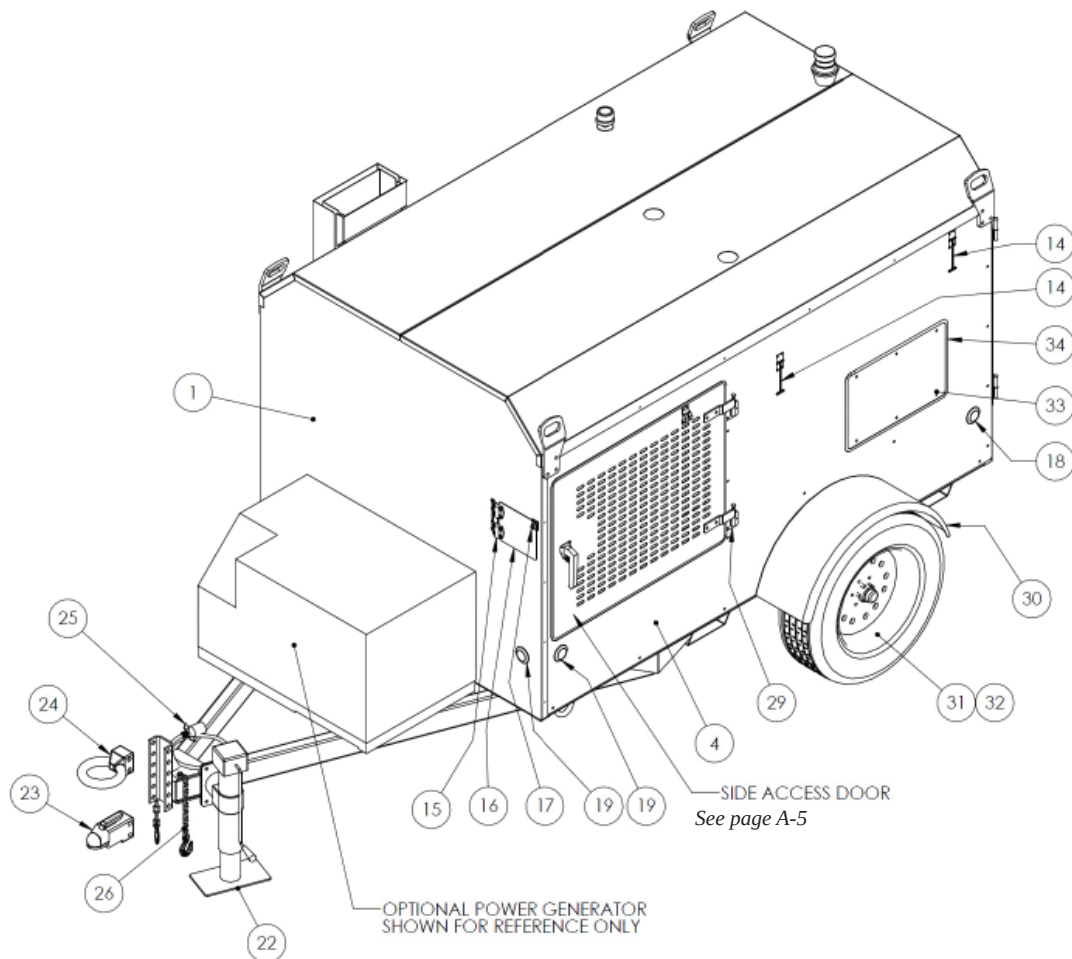
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Trailer Exterior Hardware, HK 300



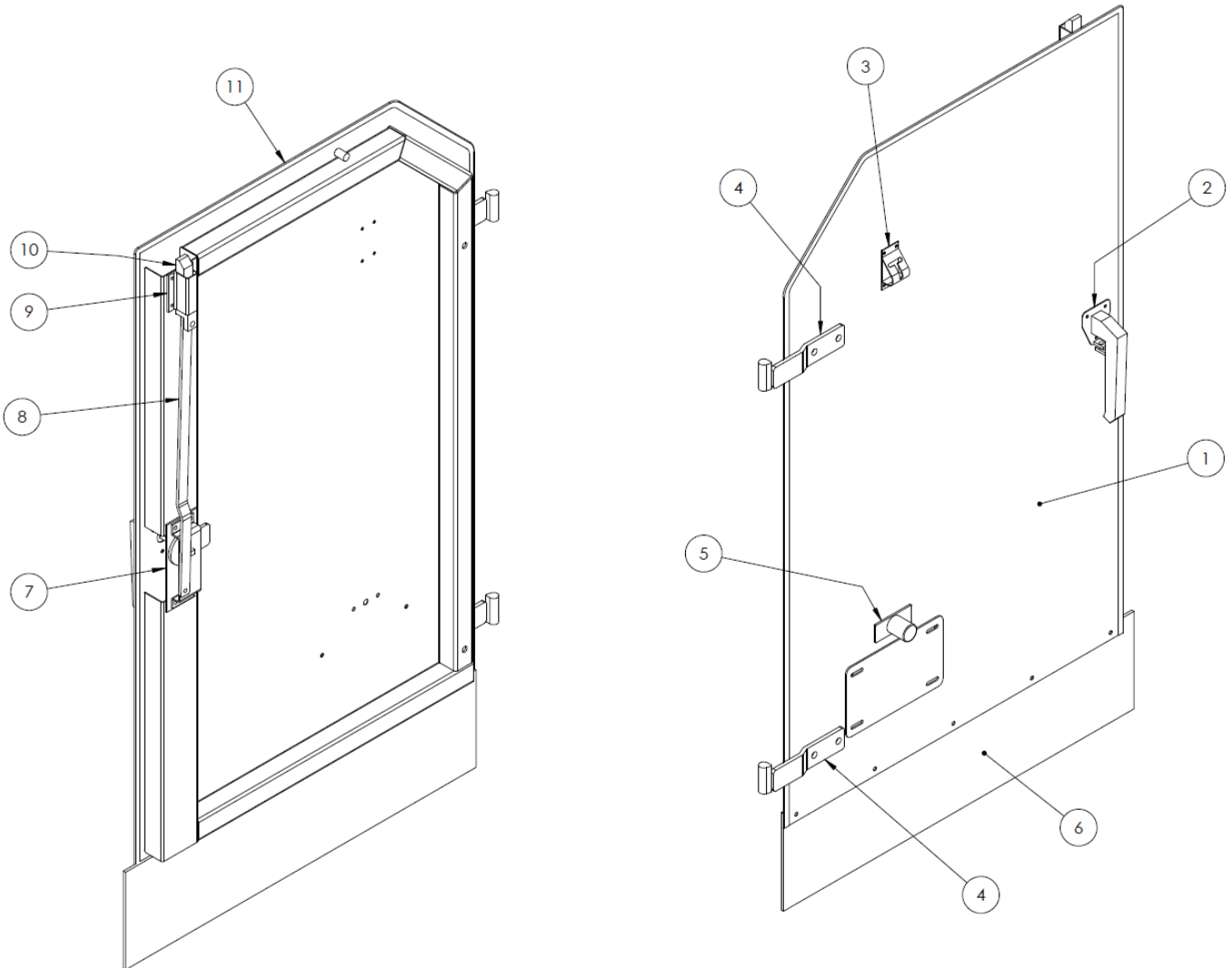
Ref.	Part	Description	Qty.
1	448097	PANEL, FRONT	1
2	448170	PANEL, ROOF	1
3	448133	PANEL, ROOF	1
4	448171	PANEL, SIDE	1
5	448134	PANEL, SIDE	1
6	448204	HOOK, LIFT, RHS	2
7	448206	HOOK, LIFT, LHS	2
8	451111	EXHAUST, GUARD	1
9	451110	EXHAUST, TERMINAL	1
10	451113	EXHAUST, TERMINAL	1
11	451112	CAP, EXHAUST	1
12	445165	FUEL TANK BREATHER	1
13	446707	STROBE LIGHT	1
14	448009	DOOR KEEPER , 6"	3
15	448248	HINGE	2
16	448243	FUEL DOOR	1
17	451152	FUEL DOOR HASP	1

Trailer Exterior Hardware HK 300 - Cont'd



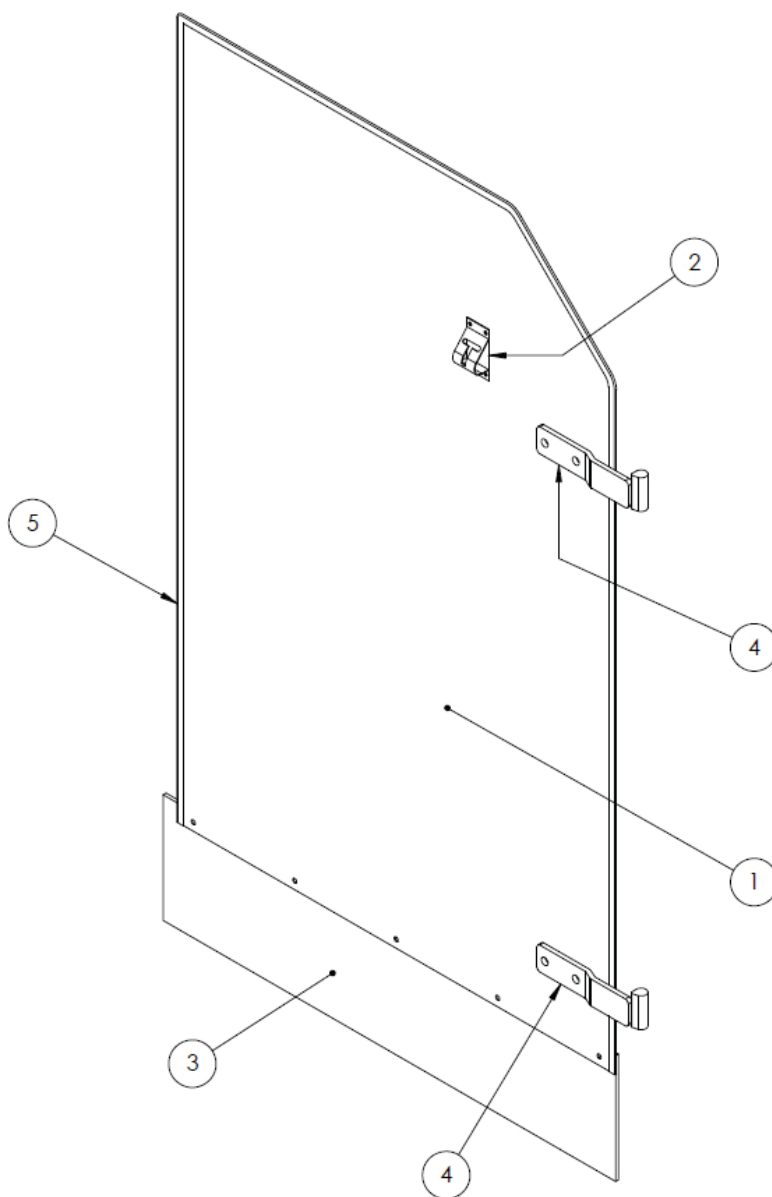
Ref.	Part	Description	Qty.
18	477008	LAMP, RED, CLEARANCE 2"	2
19	477007	LAMP, ORANGE, CLEARANCE 2"	4
20	477009	LAMP, STOP	2
21	448021	LIGHT, LICENCE PLATE, HK	2
22	448210	TRAILER JACK, 8000 LB	1
23	448003	BALL COUPLER, 2-5/16"	1
24	449025	PINTLE COUPLER	1
25	448017	CABLE, ARTIE RV w/7 Pin CONNECTOR	1
26	448016	CHAIN, SAFETY 3/8" x 30" GR 43	2
27	448213	REAR DOOR HINGE ASSY. - RH	1
28	448212	REAR DOOR HINGE ASSY. - LH	1
29	448191	SIDE DOOR HINGE	2
30	448799	FENDER, 10.75 x 36	2
31	448001	RIM AND TIRE ASSY. (225-75-R15)	2
32	448000	AXLE ASSY.	1
33	447380	SERVICE PANEL	1
34	447160	SEAL, BULB	1

Rear Door Assembly - Left Hand, HK 300



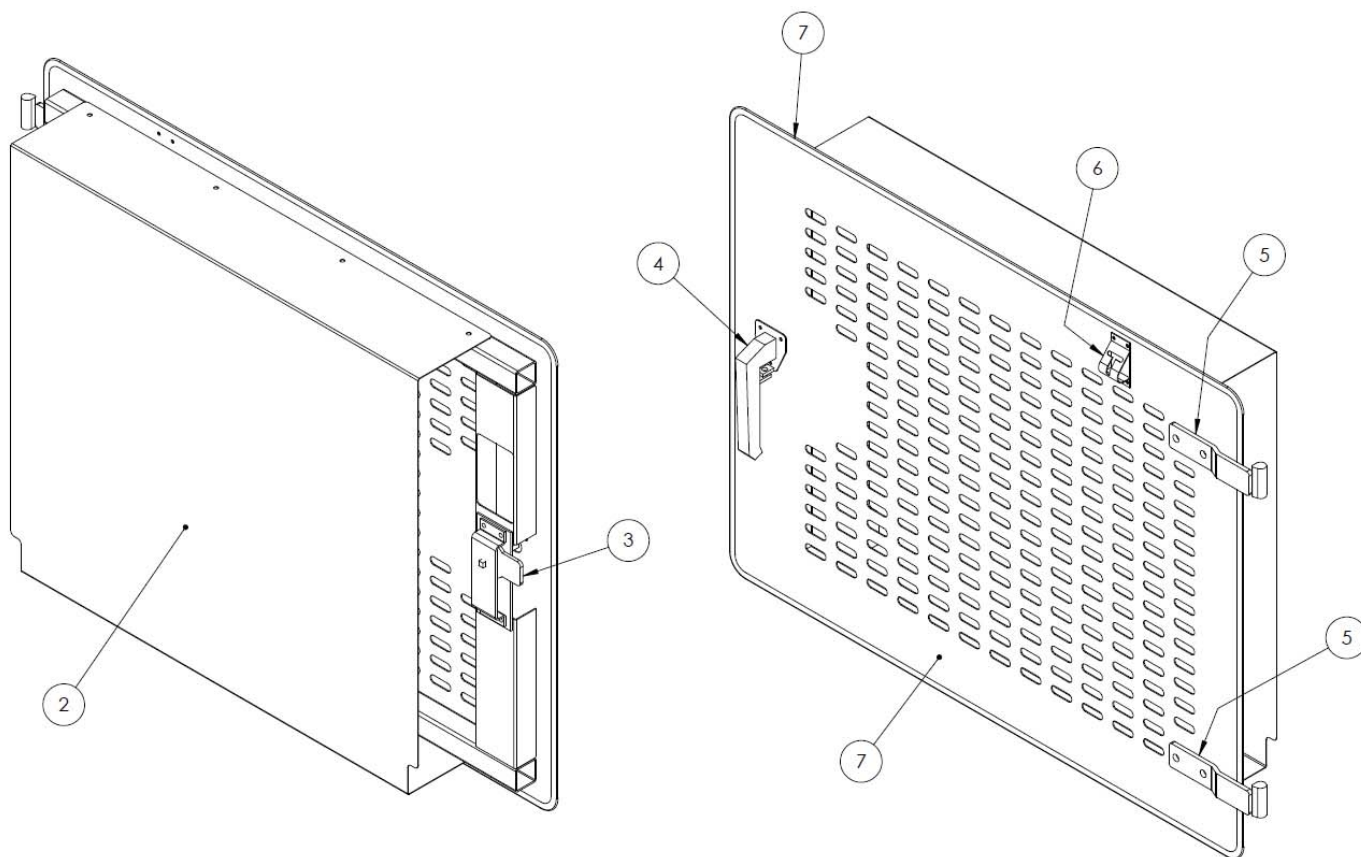
Ref.	Part	Description	Qty.
1	448150	445155 DOOR REAR, LH	1
2	448006	HANDLE, DOOR	1
3	448009	KEEPER, DOOR , 6 IN	1
4	448178	SIDE DOOR HINGE ASSY.	2
5	448021	LIGHT, LICENCE PLATE	1
6	448011	FLAP, 8 IN, REAR DOOR	1
7	448007	LATCH, 3 POINT, CENTER CASE	1
8	448093	ROD	1
9	448091	CASE BOLT	1
10	448092	BOLT	1
11	447160	SEAL, BULB	1

Rear Door Assembly - Right Hand, HK 300



Ref.	Part	Description	Qty.
1	448156	REAR DOOR, RIGHT	1
2	448009	KEEPER, DOOR , 6 IN	1
3	448011	FLAP, 8 IN, REAR DOOR	1
4	448178	SIDE DOOR HINGE ASSY.	2
5	447160	SEAL, BULB	1

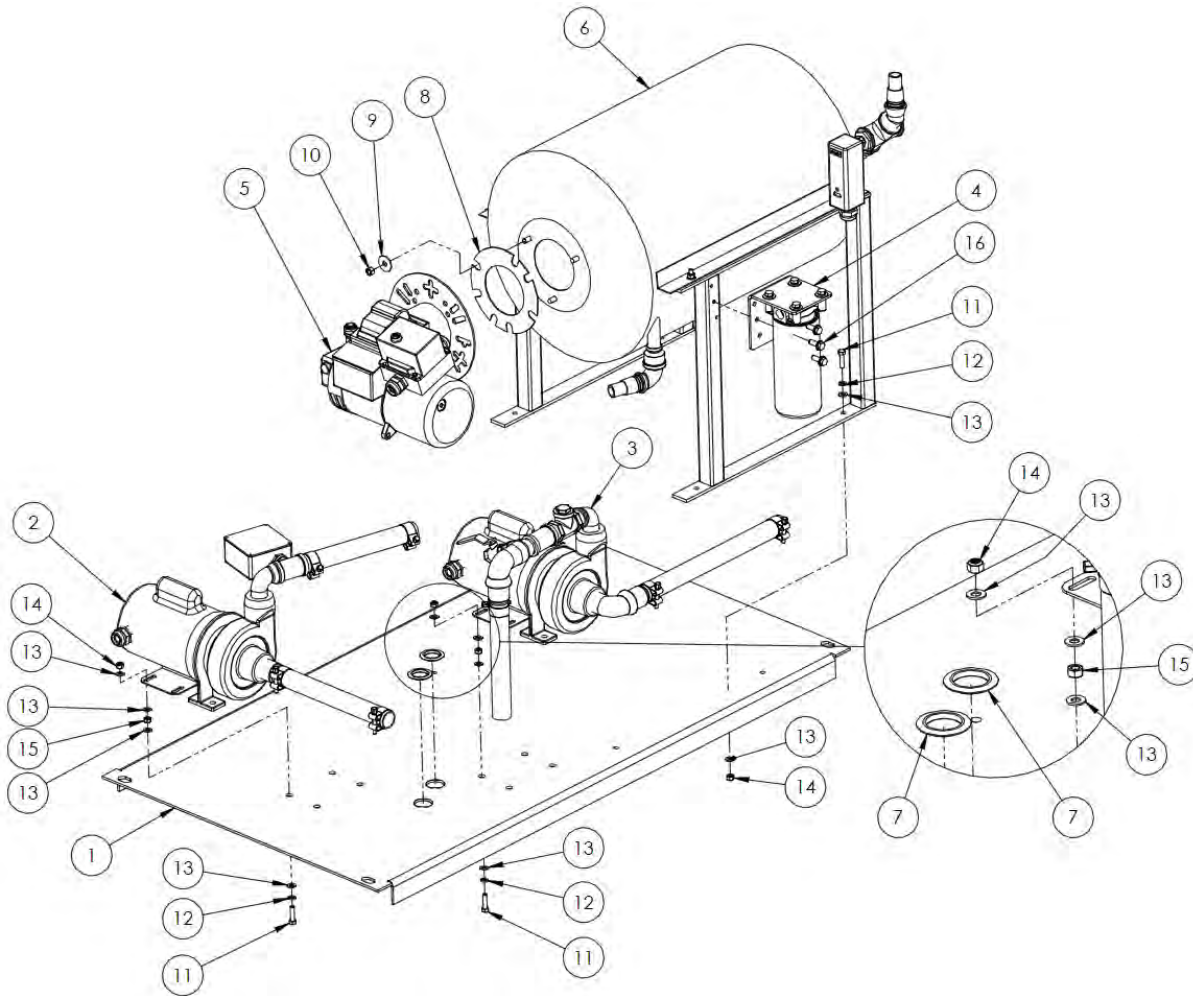
Side Door Assembly, HK 300



Ref.	Part	Description	Qty.
-	448157	COMPLETE SIDE DOOR ASSY	1
1	448136	PANEL, SIDE DOOR	1
2	448142	GUARD WIND SHIELD	1
3	448007	LATCH, 3 POINT, CENTER CASE	1
4	448006	HANDLE, DOOR	1
5	448178	SIDE DOOR HINGE ASSY.	2
6	448009	KEEPER, DOOR , 6 IN	1
7	447160	SEAL, BULB	1

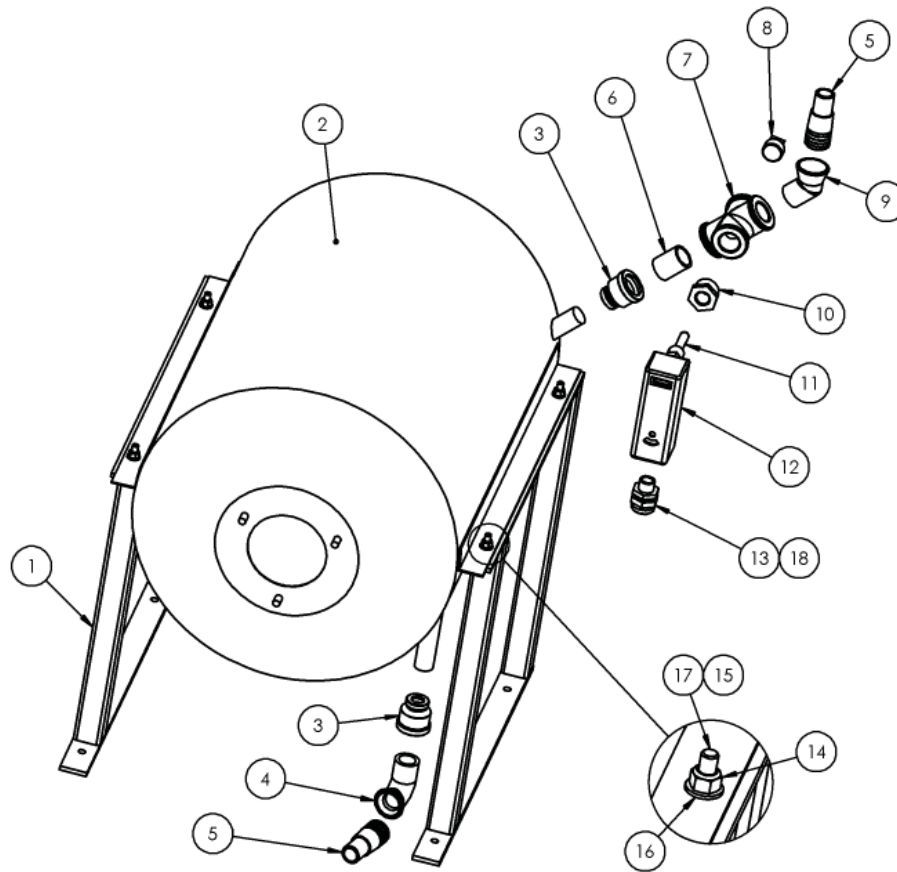
Service Notes

Pump Plate Assembly, HK 300



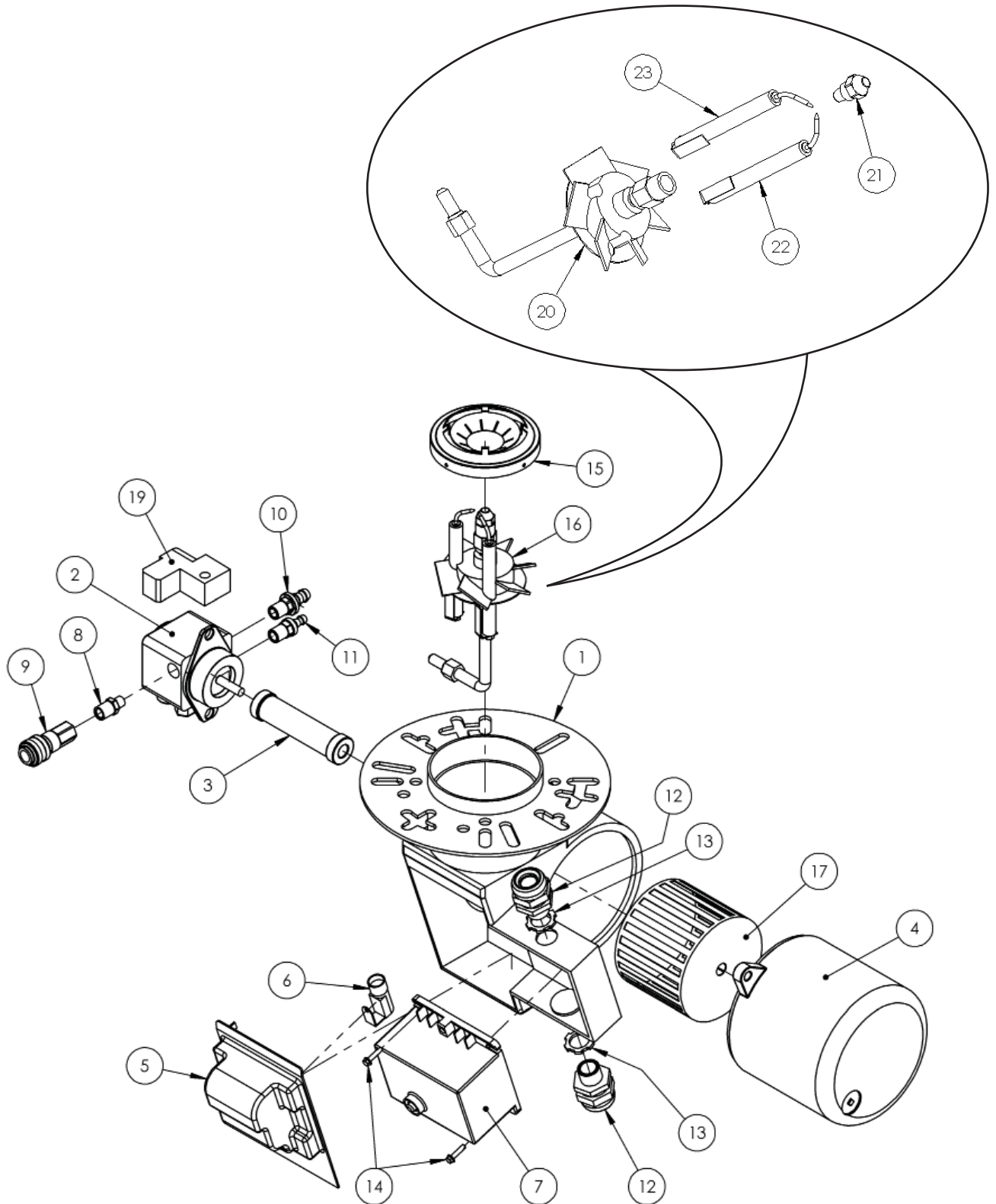
Ref.	Part	Description	Qty.
1	448103	PLATE, PUMP FLOOR, HK 300	1
2	See Page B-5	CIRCULATION PUMP ASSEMBLY, HK 300	1
3	See Page B-6	FIELD PUMP ASSEMBLY, HK 300	1
4	See Page B-8	FUEL FILTER ASSEMBLY, HK 300	1
5	451320	BURNER ASSEMBLY (See Page B-3)	1
6	See Page B-2	HEATING COIL ASSEMBLY, HK 300	1
7	440203	BUSHING, SNAP-IN	2
8	441136	SEAL, GASKET	1
9	174987	WASHER, 3/8 FLAT, WIDE	3
10	227830	NUT, HEX, 3/8	3
11	225990	BOLT, HEX HEAD, 5/16 X 1-1/4	12
12	176220	WASHER, 5/16, SPRING	12
13	174990	WASHER, FLAT, 5/16	40
14	227831	NUT, HEX, NYLON-INSERT, 5/16	12
15	227830	NUT, HEX, 3/8	8
16	228061	BOLT, FLANGE HEAD, SELF-DRILLING, 3/8 X 1	3

Heating Coil Assembly, HK 300

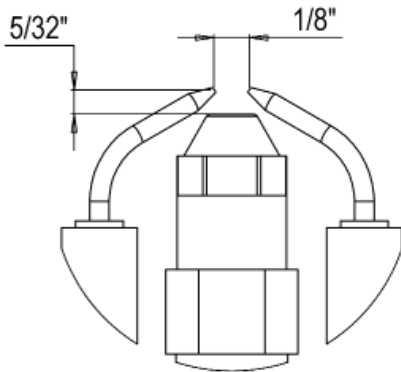


Ref.	Part	Description	Qty.
1	451060	STAND FRAME ASSEMBLY	2
2	448055	ASSEMBLY, COIL,	1
3	444816	BELL REDUCER, 1 x 1/2" GLVD	2
4	444820	ELBOW, 1" 90D STREET GLVD	1
5	444822	1" GALV, HOSE BARB	2
6	444740	1" GALVANIZED CLOSE NIPPLE	1
7	444850	CROSS, 1 x 1 x 1 GLVD	1
8	444866	PLUG, 1" NPT GLVD	1
9	446827	STREET ELBOW 1" 45 DEGREE GLVD	1
10	444817	REDUCER BUSHING, 1" x 1/2" GLVD	1
11	446543	HIGH LIMIT AQUASTAT WELL	1
12	446542	AQUASTAT, HIGH LIMIT HONELYWELL	1
13	446260	CONNECTOR, 1/2" STRESS RELIEF	1
14	225990	BOLT, HEX HEAD, 5/16 x 1-1/4	8
15	174990	WASHER, 5/16	4
16	176220	WASHER, 5/16, SPRING	4
17	227831	NUT, HEX, NYLON-INSERT 5/16	4
18	112510	NUT, LOCK CONDUIT 1/2	1

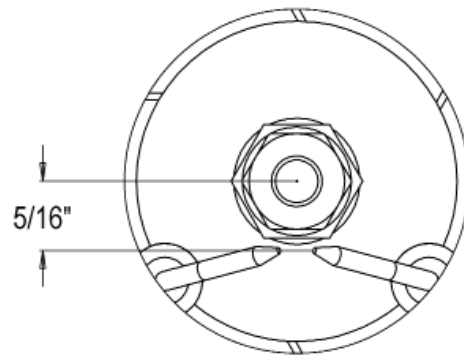
Burner Assembly, HK 300



Burner Assembly, HK 300 - Cont'd



PARTIAL SIDE VIEW

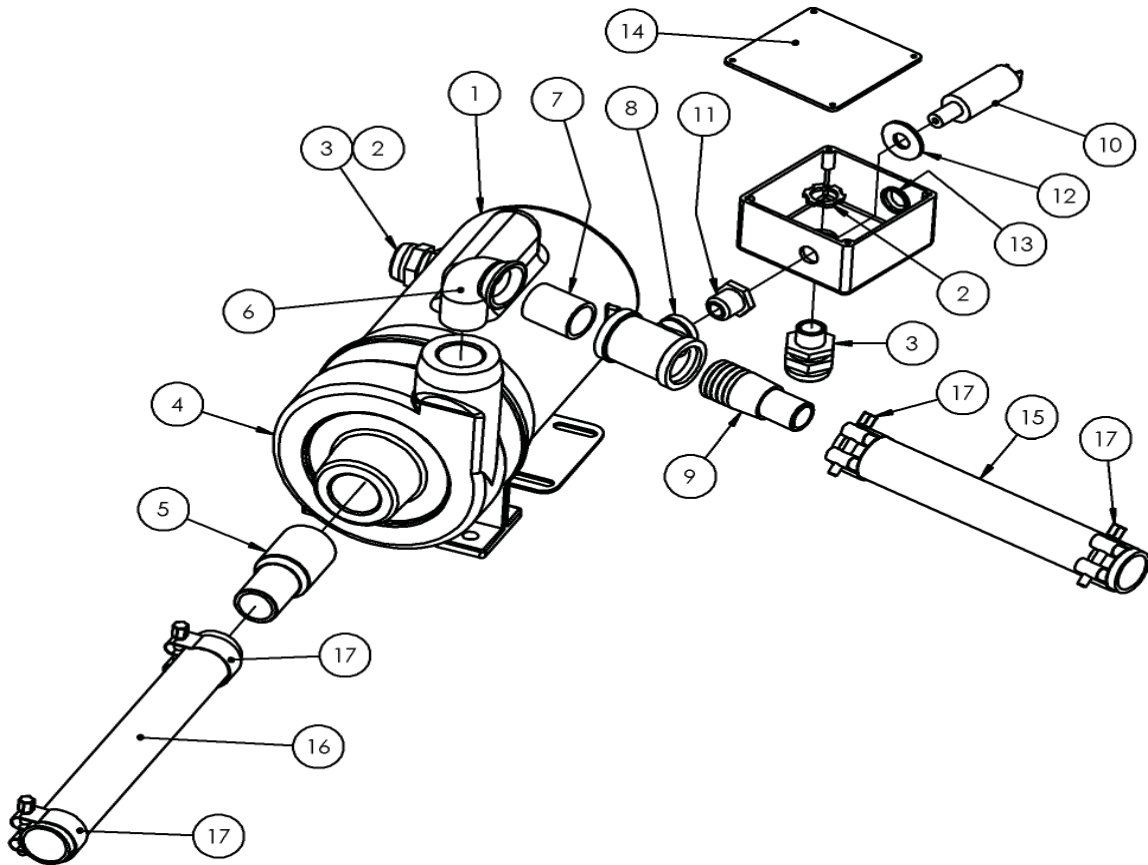


PARTIAL TOP VIEW

SETUP AT INSTALLATION OF ITEM 22 AND 23

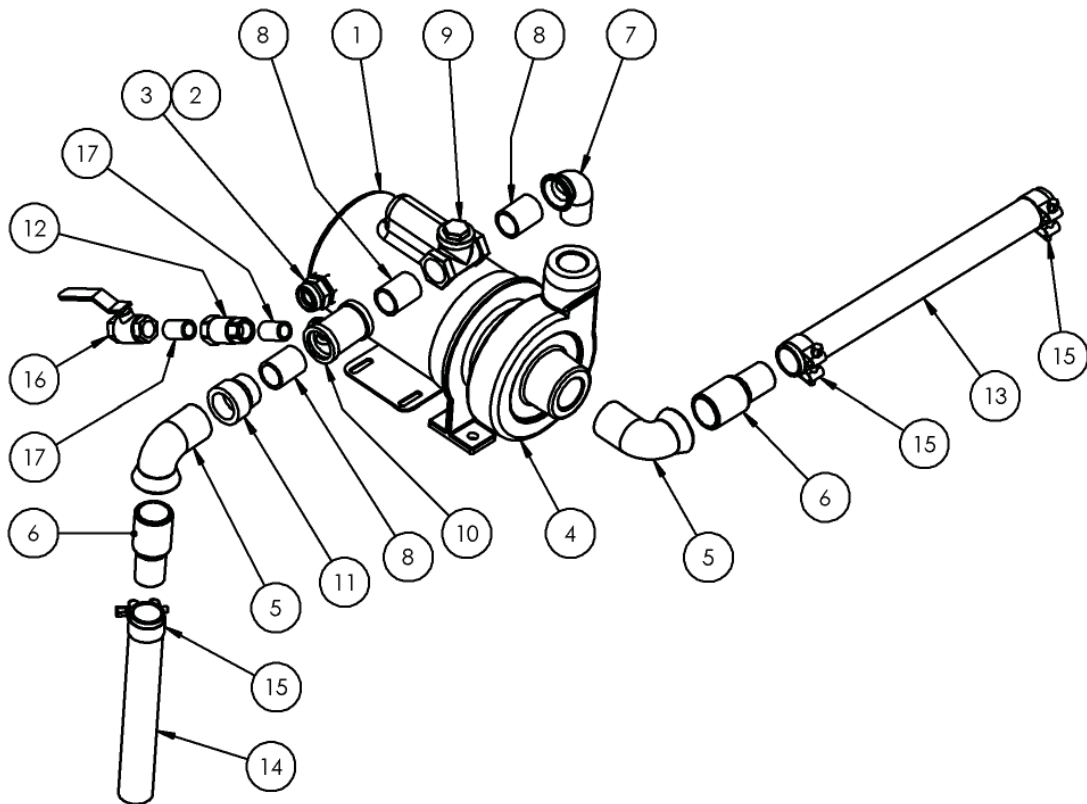
Ref.	Part	Description	Qty.
1	449048	BURNER CHASSIS	1
2	449001	BURNER, FUEL PUMP 3450 RPM, 3 GPH	1
3	449032	BURNER, FLEXIBLE COUPLING	1
4	449024	BURNER, FAN MOTOR	1
5	449014	TRANSFORMER, IGNITOR	1
6	446491	SENSOR, CAD CELL LEAD	1
7	446490	SENSOR, PRIMARY CONTROL RELAY	1
8	444814	ADAPTER, 1/8 NPT x 1/4" MALE	1
9	444032	DISCONNECT, 1/4" FEMALE QUICK	1
10	444021	NIPPLE, 1/4 NPT x 3/8" BRSS HOSE	1
11	444042	NIPPLE, 1/4 NPT x 1/4" BRSS HOSE	1
12	446260	CONNECTOR, 1/2" STRESS RELIEF	2
13	112510	NUT, LOCK CONDUIT 1/2	2
14	162830	8-18 X 3/4 HWH SCREW	2
15	446766	BURNER, RETENTION HEAD	1
16	446615	BURNER, DRAWER ASSEMBLY	1
17	449028	BURNER, FAN SQUIRREL CAGE	1
18	446768	BURNER, AIR BAND	1
19	446794	SOLENOID, FUEL	1
20	446615	BURNER DRAWER, ALKOTA	1
21	449212	NOZZLE, FUEL, 1.65 GPH 80D A	1
21*	441138	NOZZLE, FUEL, 1.50 GPH 80D A (High Altitude)	1
22	449003	BURNER, ELECTRODE RHS	1
23	449030	BURNER, ELECTRODE LHS	1

Circulation Pump, HK 300



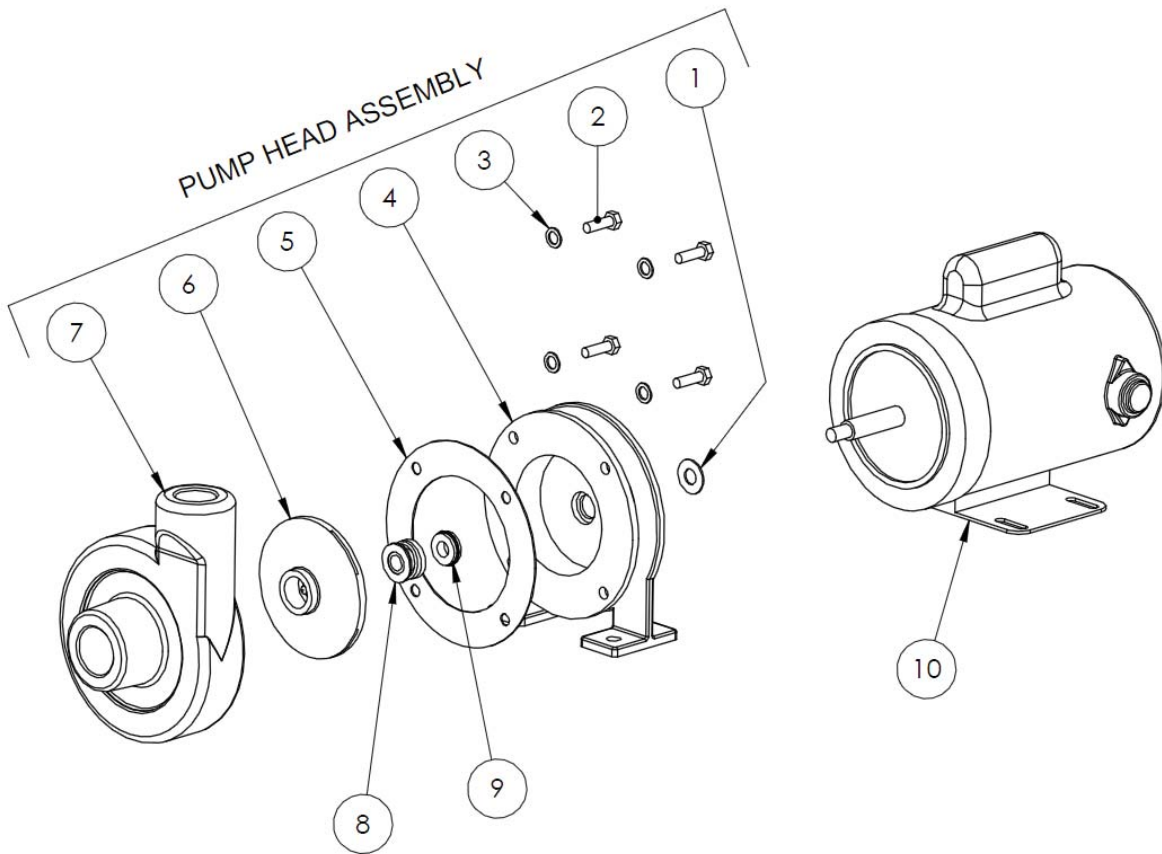
Ref.	Part	Description	Qty.
1	219110	MOTOR, 1/2HP 115/230V	1
2	112510	NUT, LOCK CONDUIT 1/2	2
3	446260	CONNECTOR, 1/2" STRESS RELIEF	2
4	448045	PUMPHEAD, 1/2 HP (See Page B-7)	1
5	444330	HOSE BARB, 1-1/4" GLVD	1
6	444820	ELBOW, 1" 90D STREET GLVD	1
7	444740	1" GALVANIZED CLOSE NIPPLE	1
8	444821	1" x 1" x 1/2" GALVANIZED TEE	1
9	444822	1" GALV, HOSE BARB	1
10	447011	SWITCH, PRESSURE	1
11	444832	BUSHING, 1/2" x 1/4" GLVD	1
12	446613	WASHER, SMALL PRESSURE SWITCH	1
13	477045 Bottom	JUNCTION BOX	1
14	477045 Top	JUNCTION BOX, COVER	1
15	442156	HOSE, STREAM , 1" x 12" LG.	1
16	442162	HOSE, WATER SUCTION, 1-1/4" x 12" LG.	1
17	444810	T-BOLT, 1" CLAMP	4

Field Pump, HK 300



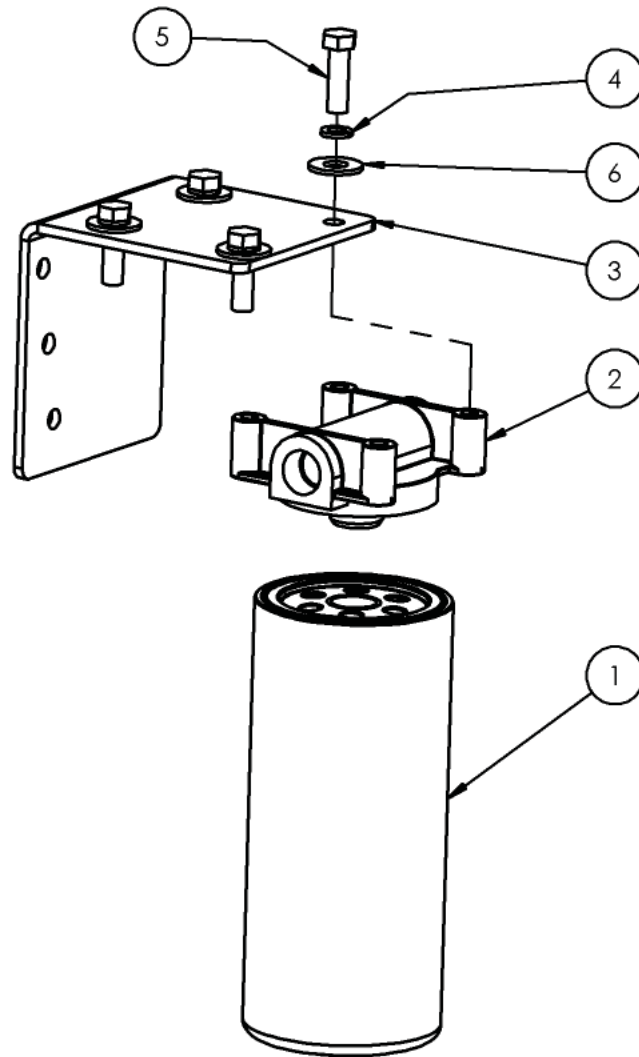
Ref.	Part	Description	Qty.
1	219505	MOTOR, 1 HP 115/230V	1
2	446260	CONNECTOR, 1/2" STRESS RELIEF	1
3	112510	NUT, LOCK CONDUIT 1/2	1
4	448054	PUMPHEAD, 1 HP (See Page B-7)	1
5	444250	ELBOW, STREET 1 1/4" 90D GLVD	2
6	444330	HOSE BARB, 1-1/4" GLVD	2
7	444820	ELBOW, 1" 90D STREET GLVD	1
8	444740	1" GALVANIZED CLOSE NIPPLE	3
9	444811	1" BRASS SWING CHECK VALVE	1
10	444821	1" x 1" x 1/2" GALVANIZED TEE	1
11	446800	BUSHING, REDUCER 1-1/4 x 1	1
12	444731	VALVE, 1/2" BRASS IN-LINE CHECK	1
13	442162	HOSE, 1-1/4", SUCTION	1
14	446815	HOSE, 1-1/4", 150 PSI	1
15	444810	T-BOLT, 1" CLAMP	3
16	444120	VALVE, BALL 1/2"	1
17	444440	NIPPLE, 1/2" CLOSE GLVD	2

Pump Assembly Breakdown, HK300



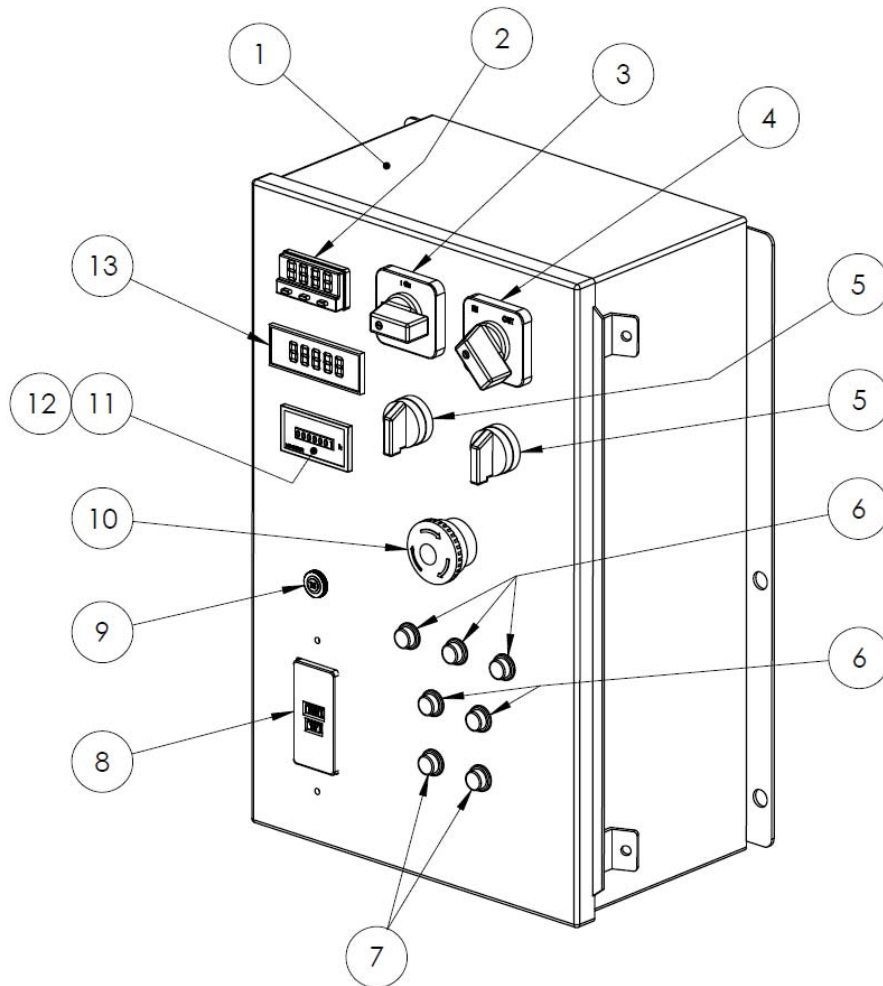
Ref.	Part	Description	Qty.
--	448054 **	PUMP HEAD ASSEMBLY,	1
1	449002	FINGER SEAL	1
2	225980	BOLT, 5/16 UNC	4
3	175500	WASHER, 5/16	4
4	---	PUMP BODY, MOTOR SIDE	1
5	449012	GASKET, PUMP	1
6	---	PUMP Impeller	1
7	---	PUMP BODY, CAVITY SIDE	1
8	---	SEAL, MOTOR SIDE	1
9	---	SPRING SEAL	1
10	219505	MOTOR, 1 HP 1-115/230V	1
	449059	SEAL KIT (Includes items: 8, and 9)	
	**	PUMP HEAD ASSEMBLY (Includes items: 1 to 7)	

Filter Assembly, HK300



Ref.	Part	Description	Qty.
1	445157	FUEL FILTER	1
2	445156	FUEL FILTER HEAD	1
3	446821	BRACKET, FUEL FILTER	1
4	176220	WASHER, 5/16, SPRING	4
5	226220	BOLT, HEX HEAD 3/8 UNC	4
6	174986	WASHER, 3/8	4

Control Box Assembly, HK 300

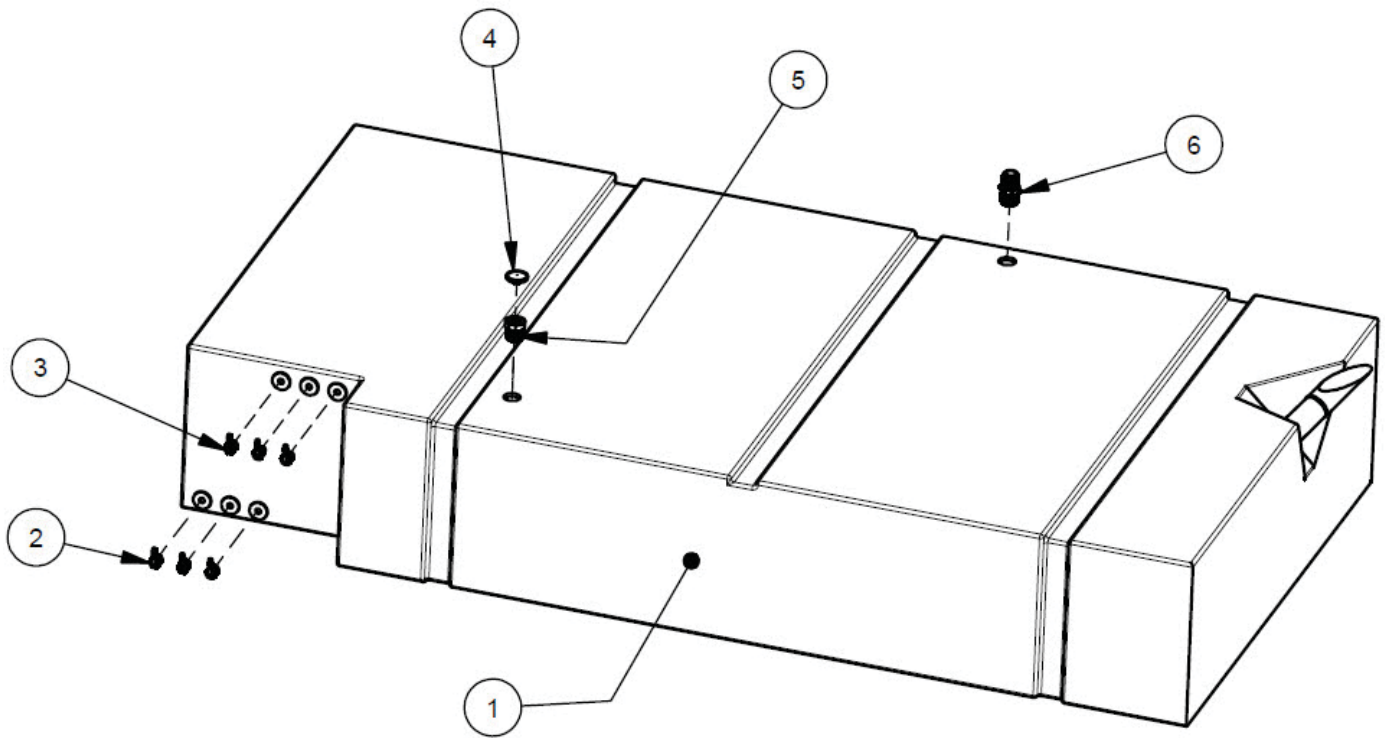


Ref.	Part	Description	Qty.
1	451350	BOX, ELECTRICAL	1
2	450093	TEMPERATURE CONTROLLER	1
3	446513	SWITCH, CIRCUIT POWER	1
4	446514	SWITCH, REEL DIRECTION	1
5	446517	SWITCH	2
6	446130	LIGHT, INDICATOR, AMBER	5
7	446140	LIGHT, INDICATOR, RED	2
8	446110	GROUND FAULT	1
9	446010	CIRCUIT BREAKER, 15 AMP	1
10	446527	SWITCH, EMERGENCY STOP	1
11	446120	HOUR METER	1
12	446125	HOUR METER, FACE PLATE	1
13	446519	RETURN READOUT	1

Service Notes

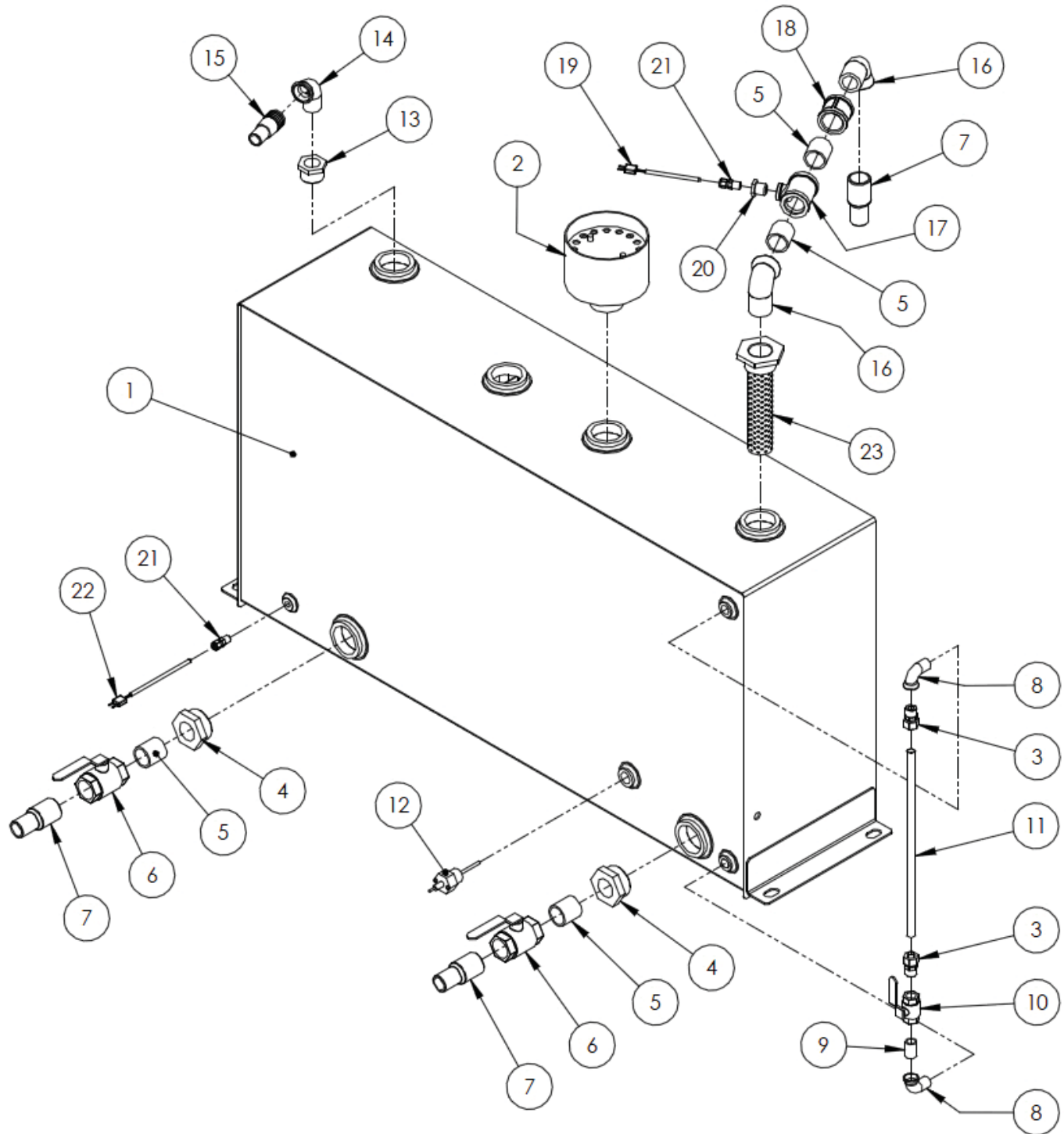
Service Notes

Fuel Tank Assembly, HK 500



Ref.	Part	Description	Qty.
1	447400	FUEL TANK	1
2	444006	1/4" NPT X 1/4" 90 DEG.	1
3	444007	1/4" NPT X 3/8" 90 DEG.	1
4	447396	GUAGE	1
5	447397	GROMMET	1
6	444350	NIPPLE, 1" NPT x 1" HOSE BARB, PVC	1

Glycol Tank Assembly, HK 300

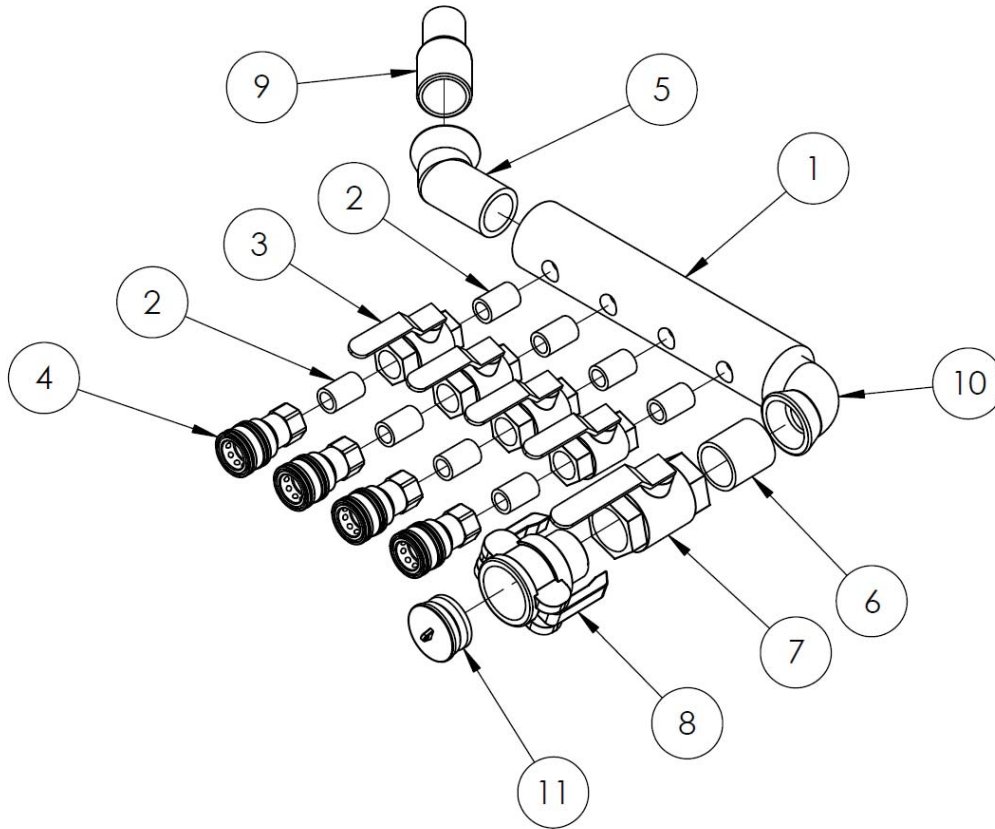


Glycol Tank Assembly, HK 300 - Cont'd

Ref.	Part	Description	Qty.
1	451050	GLYCOL TANK WELDMENT	1
2	444704	BREATHER, 500MBTU ASSY	1
3	444200	COMPRESSION, 1/2NPT X 5/8"	2
4	444140	BUSHING, 2 X 1 1/4" GLVD	2
5	444480	CLOSE NIPPLE, 1-1/4, GLVD	4
6	444110	BALL VALVE, 1-1/4	2
7	444330	HOSE BARB, 1-1/4, GLVD	3
8	444610	ELBOW, STREET 1/2" GLVD	2
9	444440	NIPPLE, 1/2 CLOSE, GLVD	1
10	444120	VALVE, BALL 1/2"	1
11	446756	POLY TUBING, 1/2ID X 5/8"OD	1
12	441140	LOW LEVEL CUT OFF - PROBE	1
13	444818	REDUCER, 1 1/4x1" BUSHING GLVD	1
14	444820	ELBOW, 1" 90D STREET GLVD	1
15	444822	1" GALV, HOSE BARB	1
16	444250	ELBOW, STREET 1-1/4 X 90DEG, GLVD	2
17	444540	TEE, REDUCER 1 1/4X1 1/4X1/2"	1
18	444807	COUPLING, 1 1/4" GLVD	1
19*	446518	RETURN THERMOCOUPLE ASSY	1
20	444832	BUSHING, 1/2"x1/4" GLVD	1
21	444758	FITTING, COMPRESSION	2
22*	446704	GLYCOL TANK THERMOCOUPLE ASSY.	1
23	445846	RETURN STRAINER	1

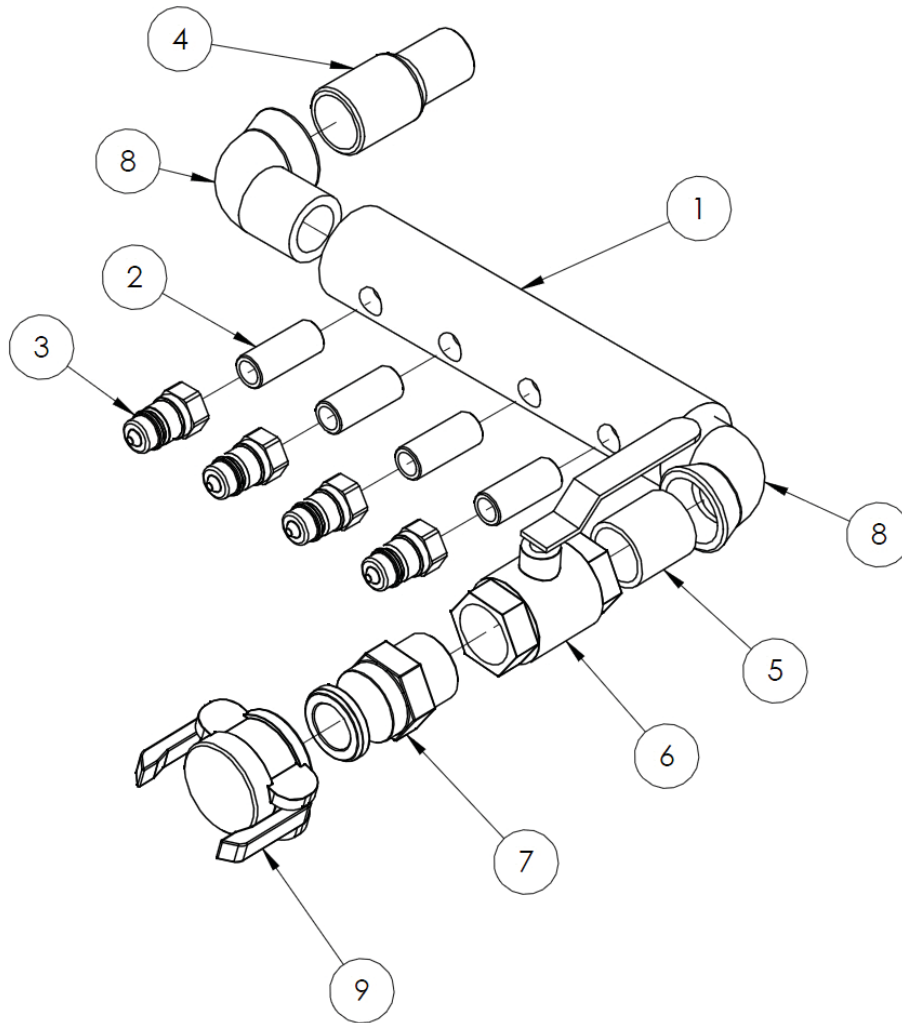
* Order this item with Compression Fitting - Item 21

Supply Manifold, HK 300



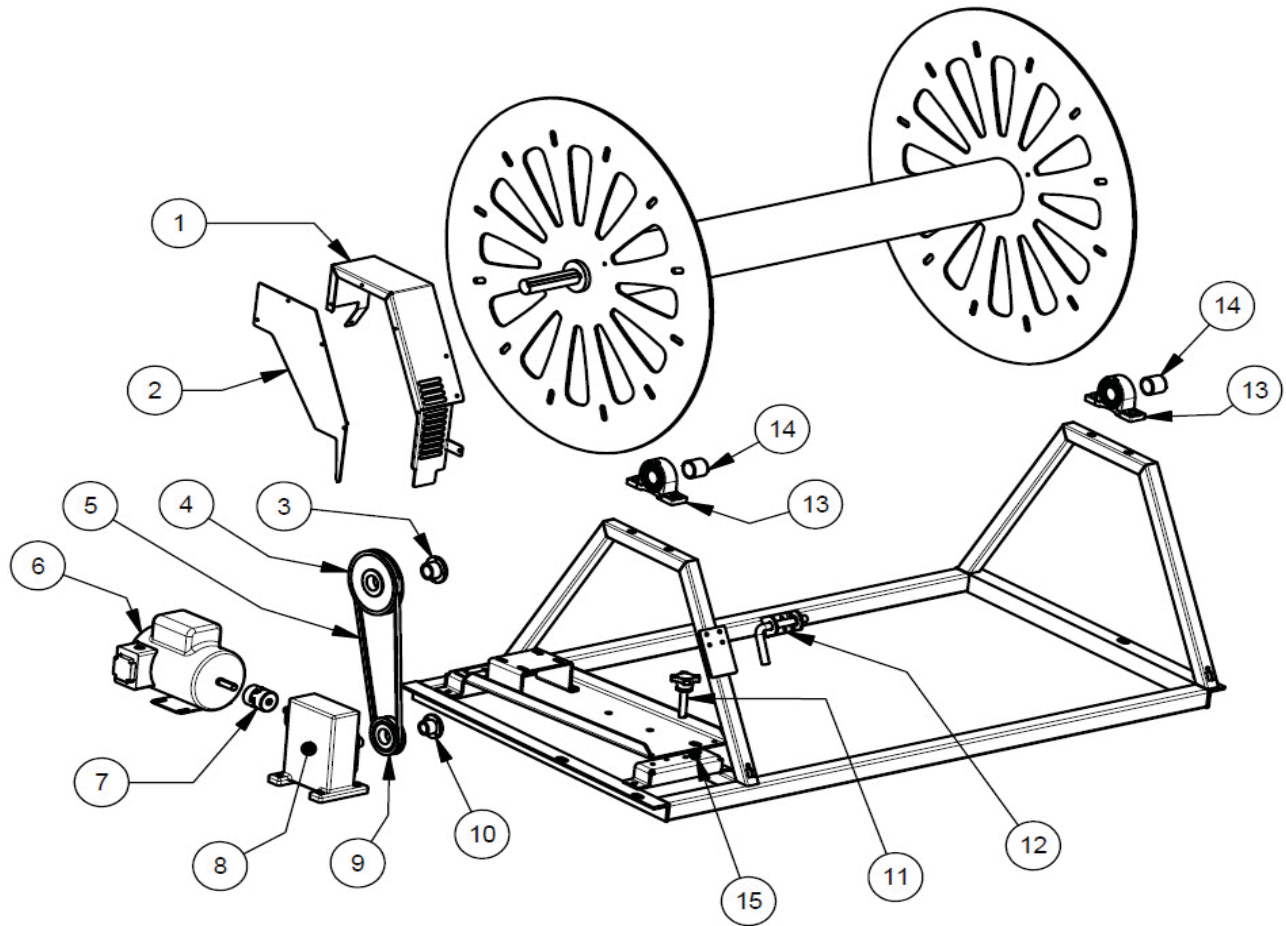
Ref.	Part	Description	Qty.
1	451200	MANIFOLD, 4 PLACE	1
2	444440	NIPPLE, 1/2 CLOSE, GLVD	8
3	444081	VALVE, BALL 3/4 NPT, BRASS	4
4	444901	QUICK CONNECT, 1/2, FEMALE	4
5	444750	ELBOW, 1-1/4 X 45 DEG STREET, GLVD	1
6	444480	CLOSE NIPPLE, 1-1/4, GLVD	1
7	444110	VALVE, BALL 1-1/4	1
8	444751	CAMLOCK, 1-1/4 NPT MAL X 1-1/4 FEM	1
9	444330	HOSE BARB, 1-1/4, GLVD	1
10	444250	ELBOW, STREET 1-1/4 X 90 DEG, GLVD	1
11	444086	PLUG, 1-1/4, CAMLOCK, MALE	1

Return Manifold, HK 300



Ref.	Part	Description	Qty.
1	451200	MANIFOLD, 4 PLACE HK150	1
2	444805	NIPPLE, 1/2 X 2, GLVD	4
3	444900	QUICK CONNECT, 1/2, MALE	4
4	444330	HOSE BARB, 1-1/4, GLVD	1
5	444480	CLOSE NIPPLE, 1-1/4, GLVD	1
6	444110	BALL VALVE, 1-1/4	1
7	444754	CAMLOCK, 1-1/4NPT X 1-1/4 MALE	1
8	444250	ELBOW, STREET 1-1/4 X 90DEG, GLVD	2
9	444087	PLUG, 1-1/4, CAMLOCK, FEM	1

Reel Assembly, HK 300



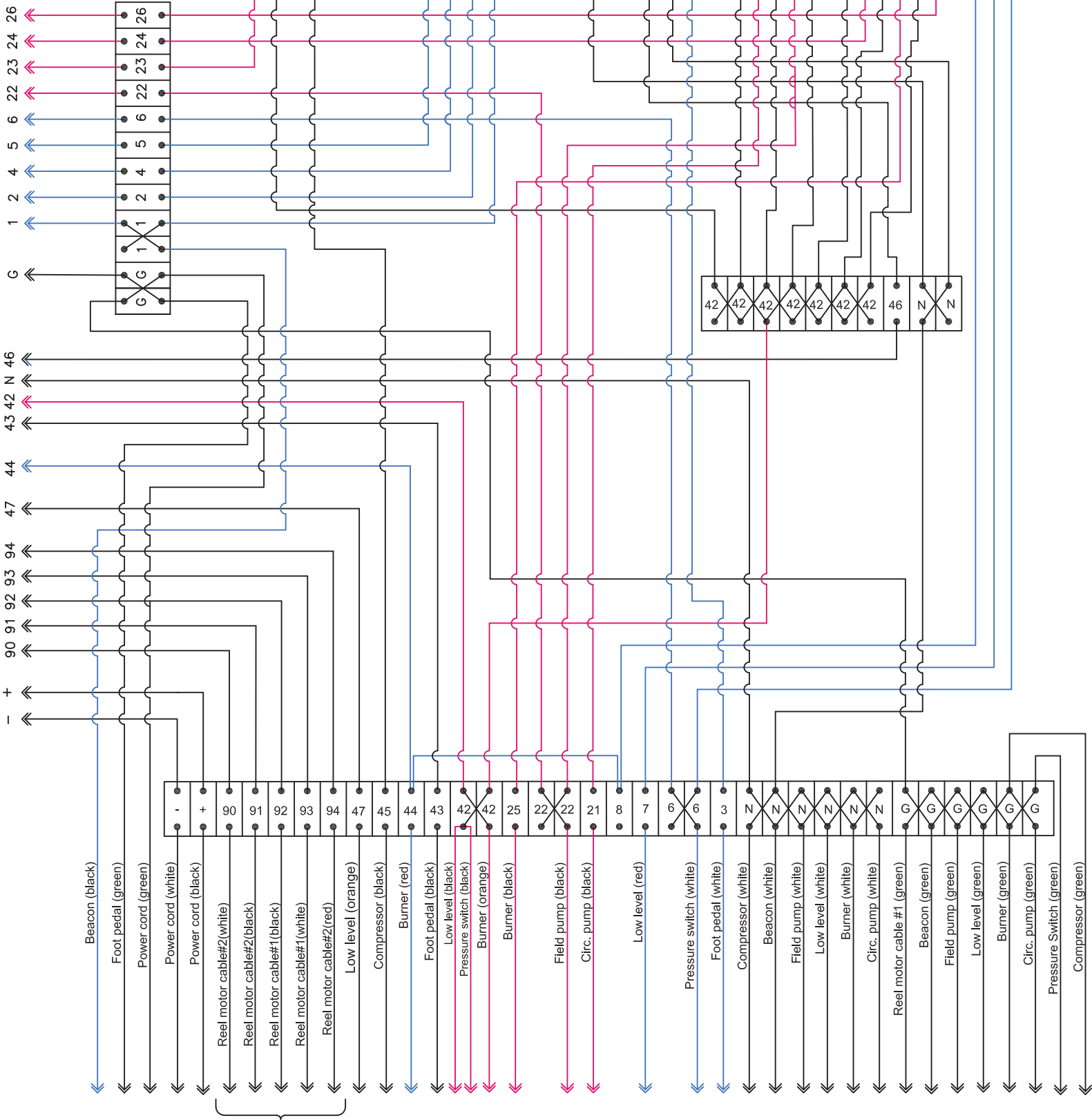
Ref.	Part	Description	Qty.
1	448232	BELT GUARD	1
2	448240	BELT GUARD, ACCESS PANEL	1
3	448263	BUSHING, REEL SHAFT	1
4	448267	PULLEY, REEL SHAFT	1
5	239592	V-BELT	1
6	219210	MOTOR, 1/2 HP	1
7	448254	COUPLING	1
8	496644	GEAR BOX	1
9	448265	PULLEY, GEAR BOX	1
10	448264	BUSHING, GEAR BOX	1
11	448256	KNOB, BELT TENSIONER	1
12*	700405	REEL LOCK PIN	1
13	451309	PILLOW BLOCK	2
14	451310	BEARING INSERT	2
15	448257	SPRING, BELT TENSIONER	1

**Please call 1 800 661 0304 for correct sizing*

Service Notes

Lined area for service notes.

TO PANEL BOX DOOR



To Reel Motor

TO COMPONENTS INSIDE TRAILER

PLC Pinout:

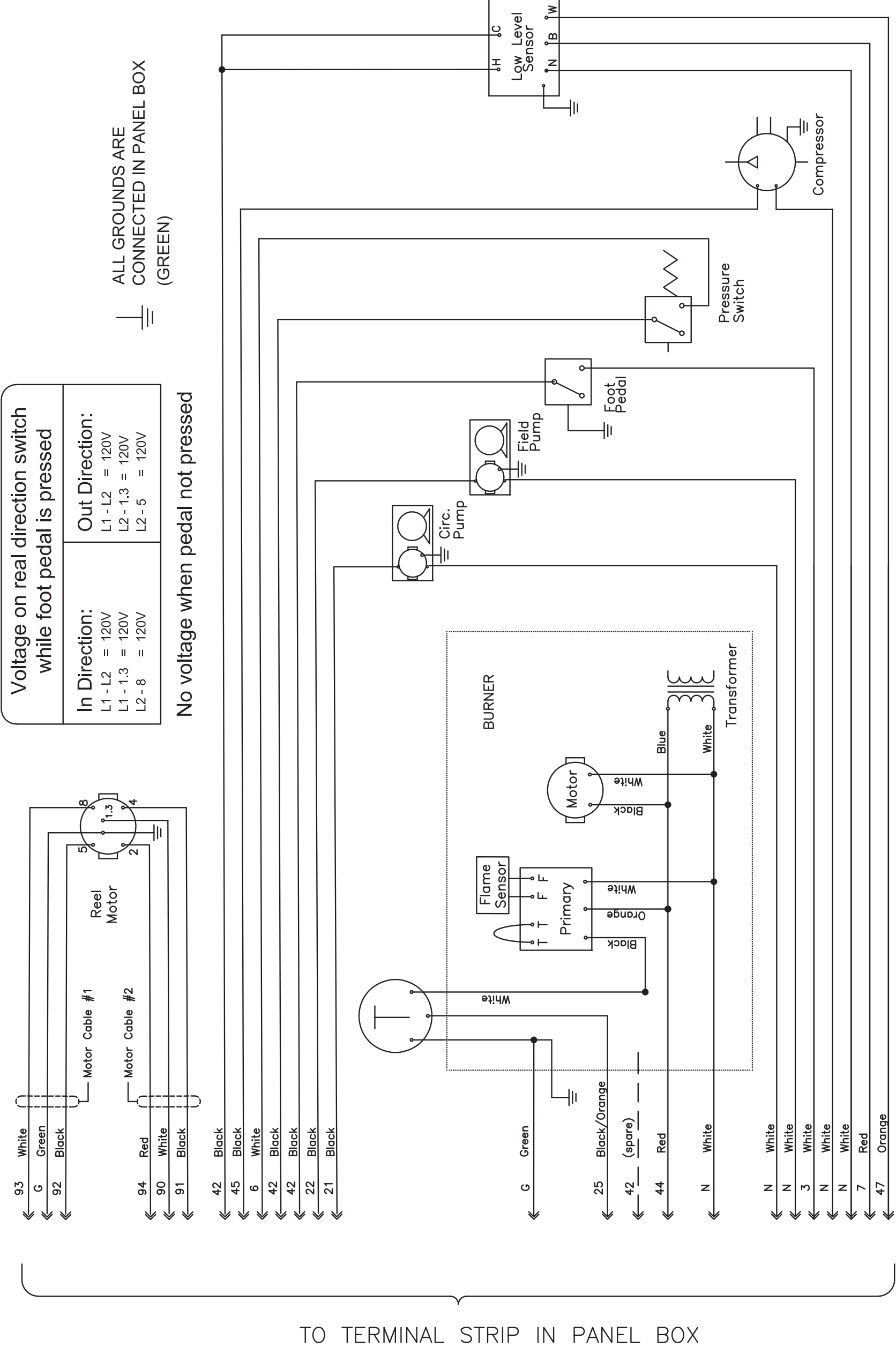
Inputs:

- I1 - Heater Switch
- I2 - Compressor Switch
- I3 - Reel, Foot Switch
- I4 - Field Switch
- I5 - Temp Switch
- I6 - Pressure Switch
- I7 - Low Glycol Sensor
- I8 - Burner Flame Sensor

Outputs:

- Q1 - Circulation Pump
- Q2 - Field Pump
- Q3 - Compressor
- Q4 - Hose Reel
- Q5 - Burner
- Q6 - Over Temp.

HK300 PANEL BOX ELECTRICAL SCHEMATIC



N

White

N

White

3

White

N

White

N

White

7

Red

47

Orange

Reel Motor

5

1.3

8

4

2

Motor Cable #1

White

Green

Black

Motor Cable #2

Red

White

Black

Flame Sensor

T

T

F

F

Primary

Black

Orange

White

Transformer

Blue

White

BURNER

Motor

Black

White

Field Pump

Ground

Pressure Switch

Ground

Compressor

Ground

Low Level Sensor

H

N

B

W

C

Ground

Foot Pedal

Ground

Circ. Pump

Ground

Field Pump

Ground

Ground

Voltage on real direction switch while foot pedal is pressed

In Direction:
L1 - L2 = 120V
L1 - 1.3 = 120V
L2 - 8 = 120V

Out Direction:
L1 - L2 = 120V
L2 - 1.3 = 120V
L2 - 5 = 120V

No voltage when pedal not pressed

ALL GROUNDS ARE CONNECTED IN PANEL BOX (GREEN)

Ground Symbol

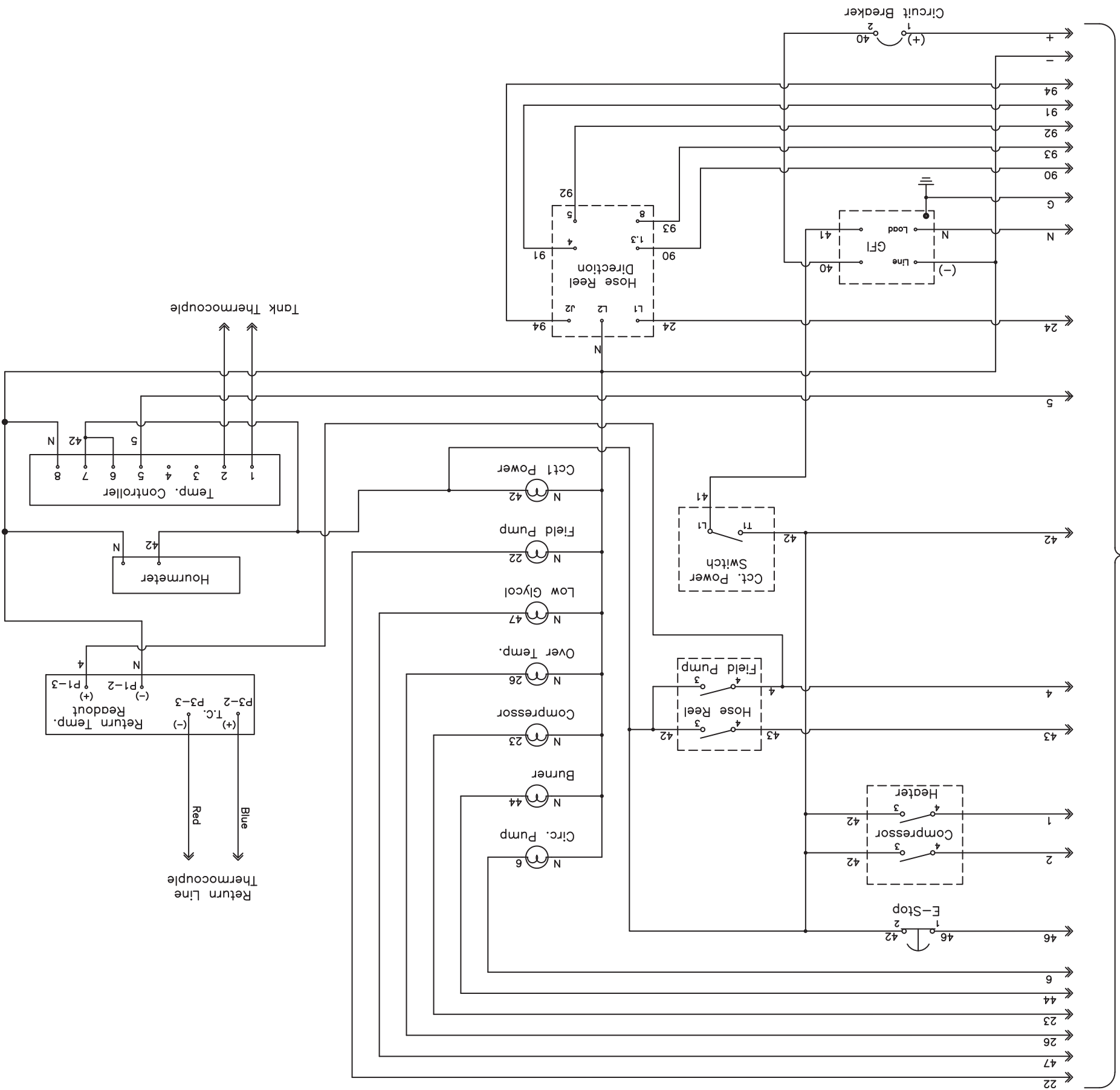
TO TERMINAL STRIP IN PANEL BOX

HK300 INSIDE TRAILER ELECTRICAL SCHEMATIC

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HK300 PANEL DOOR ELECTRICAL SCHEMATIC



TO PANEL BOX TERMINAL STRIPS



Product Brochure



Contact Us

HEAT KING

Toll Free: 1-800-661-0304
Phone: 204-885-5800
Fax: 204-885-7557

P.O. Box 234 Station L
Winnipeg, Manitoba R3H 0Z5
Canada

